Rathore and Chaurasia *Discover Sustainability* (2025) 6:1409
https://doi.org/10.1007/s43621-025-02190-w

Discover Sustainability

RESEARCH Open Access

Fine tuning large language models for hate speech detection in Hinglish and code mixed custom dataset through a socially responsible approach for safer digital platforms

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Abstract
This paper presents an extensive review and experimental framework for hate speech detection using large language models (LLMs). We explored traditional machine learning approaches and illustrated how state-of-the-art LLMs, such as BERT variants (DistilBERT and ModernBERT) and recent developments, such as DeepSeek R1, Llama 3, Gemma 3, Mistral, and Phi 3.5, enhance text classification performance. In this study, we evaluated the effectiveness of various large language models (LLMs) for Hindi–English code-mixed hate speech detection. We constructed a custom dataset using Reddit posts on social media, which were auto-labeled and subsequently verified and validated by expert annotators. LoRa adapters were used for model fine-tuning and in-domain performance evaluation of the model, with precision, recall, F1-score, and accuracy as metrics. To ensure a fair comparison, all the models were trained for the same number of epochs under the same training conditions. According to our findings, DeepSeek R1 outperformed larger general-purpose LLMs, such as Llama 3 and Gemma 3, with smaller margins. DeepSeek R1 surpassed the other models with the highest accuracy of 79 percent, demonstrating a better understanding of the complexity of code-mixed hate speech detection. These results indicate that lightweight, responsibly fine-tuned LLMs can strengthen moderation for multilingual, code-mixed communities without prohibitive computing costs. In practice, the approach offers a clear path towards safer, more inclusive platforms by pairing accuracy with bias-aware development and human oversight.

Keywords Hate speech detection, Hinglish, Large language models, Fine-tuning, Code-mixed language, Multilingual NLP

1 Introduction

The detection of hate speech has thus been categorized as an essential problem in natural language processing (NLP) as the volume of online content promoting discrimination, violence, and social concerns continues to grow. As the influence of online platforms continues to increase, the ability to automatically detect and eliminate toxic

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language has become a critical component in maintaining safe and inclusive online environments [1].

Traditionally, hate speech detection approaches have relied on classical distributed representation methods from computational linguistics in combination with keyword-based filtering, handcrafted feature engineering, and shallow machine learning techniques such as support vector machines and decision trees. Although early forms of work offered initial solutions, they did not capture all the context and semantics of offensive language, particularly in multilingual and code-mixed texts [2].

Recent advancements in Large Language Models (LLMs) have revolutionized this field. Such a method is taught through models such as BERT, GPT, and XLM-Roberta, which have emerged in recent years and perform tasks such as in-emotional semantic comprehension, context-oriented rationale, and acknowledgment of linguistic patterns. They used pre-training on large corpora and fine-tuning on task-specific data, allowing the application of classical techniques to a wide range of NLP tasks, including hate speech detection.

Emerging improvements in LLM architecture have continued to stretch the limits of the model's capability and efficiency. Revolutionary models, such as Deep Seek R1, Llama 3.2, modern BERT, distilled BERT, Mistral, and Falcon, have pushed the envelope in terms of parameter optimization, inference speed, output layering, and representational depth. These advances facilitate the more accurate detection of implicit and context-reliant hate speech, even in low-resource (code-mixed) languages. The ability of LLM to generalize across domains, languages, and platforms is particularly useful for countering hate speech at scale (e.g., in multilingual settings) [3]. These models capture subtle and complex linguistic variables along with sociolinguistic dimensions that help identify offensive content that may not be captured by shallow classification approaches. In addition, state-of-the-art training strategies (e.g., fine-tuning, zero-shot, few-shot, and multimodal) have greatly increased their success but offer distinct advantages for the endogenous diversity of online dialog [4].

Operationalizing responsibility- In this study, responsibility is concretely defined as: (i) harm mitigation via conservative confidence thresholds and mandatory human escalation for ambiguous cases; (ii) bias auditing through class balancing, shortcut/slur checks, and per-group error review; and (iii) community involvement by expert raters applying a rubric with context-aware adjudication. These steps aim to reduce over-blocking and under-detection while maintaining transparent and reproducible labeling.

This study builds on existing research by proposing an ensemble of multiple state-of-the-art LLMs in a unified framework for hate speech detection, particularly for English and Hinglish (code-mixed Hindi-English) content. We outline the literature review in Sect. 2, followed by a comprehensive methodology in Sect. 3, which includes data preprocessing techniques, model fine-tuning protocols, and the evaluation metrics used, creating a solid foundation to inspire future endeavors in the realm of multilingual and code-mixed hate speech detection. The results are disclosed in Sect. 4, after which some discussions, challenges, and limitations are presented in Sects. 5 and 6, respectively. The future scope of LLM is presented in Sect. 7, and the paper is concluded in Sect. 8. Rather than enumerating systems, our review emphasizes gaps most relevant to the Hinglish-domain shift, implicit/coded hate, and dataset quality, and positions dataset design, annotation rubric, and parameter-efficient training as targeted responses.

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1.1 Key gaps motivating this work

Despite recent advances, hate speech detection still faces pressing gaps that motivate this study, such as code-mixed and multilingual brittleness because models trained on monolingual English degrade sharply on Hinglish due to non-standard spellings, mixed scripts, and informal syntax, leading to poor generalization in real-world platforms (addressed in Sects. 3.1 and 3.2 via a curated Reddit dataset and Hinglish-aware preprocessing), and underdetection affects subtle, sarcastic, or coded expressions while context with implicit hate inflates false positives/negatives. For comparative evaluation of context-rich LLMs as well as error analysis address purely keyword-based cues specifically (Sect. 4) and Dataset quality and annotation reliability-imbalanced labels, slur-shortcut learning, and insufficient adjudication reduce robustness (addressed with a balanced dataset, hybrid auto-label plus expert adjudication pipeline, and quality controls in Sect. 3.1 and Algorithm 1). (iv) Computational efficiency-full fine-tuning of large models is costly and hinders deployment in moderation pipelines (addressed with parameter-efficient LoRA and adapters in Sect. 3.3). (v) Fair and transparent comparisons-heterogeneous tuning across models obscures true gains (addressed via common training protocol, explicit hyperparameters, and ablations in Sects. 3.4 and 4). Together, these limitations establish the urgency for a socially responsible, code-mixed-aware, and computationally efficient framework that improves reliability on real-world data.

Figure 1 summarizes how the study targets these limitations from review and selection, through Hinglish-specific preprocessing and parameter-efficient fine-tuning, to evaluation and error analysis. The funnel diagram in Fig. 1 represents the evaluation of the pipeline of this model, along with a review of existing work. The key stages of the proposed flowchart are the initial fine-tuning strategy, dataset selection and preprocessing, model optimization for Code-Mixed Hinglish text, evaluation metrics, and presentation of the performance findings. This model aims for the holistic development of LLMs that would tackle the specific linguistic and semantic difficulties in identifying hate speech in multilingual and code-mixed scenarios in an ethical and robust manner.

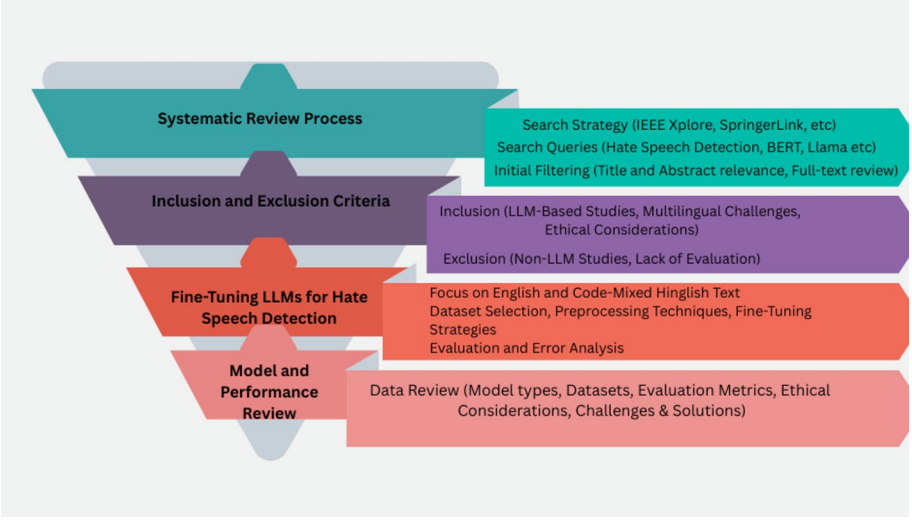

Fig. 1 Funnel diagram of fine-tuning and evaluation pipeline for hate speech detection using LLMs

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1.2 Contributions of this study

This study makes several significant contributions to the fields of hate speech detection and multilingual NLP.

1. Dataset Contribution: We developed the first comprehensive balanced dataset for Hinglish (Hindi-English code-mixed) hate speech detection, sourced from Reddit, with expert validation and human-in-the-loop filtering to handle cultural nuances and ambiguous cases.
2. Methodological Contribution- We conducted the first systematic comparative evaluation of seven state-of-the-art transformer models (DeepSeek R1, Gemma 3, LLaMA 3, DistilBERT, ModernBERT, Mistral, and Phi 3.5) for code-mixed hate speech detection using parameter-efficient low-rank adaptation fine-tuning.
3. Technical Contribution- We implemented and optimized LoRA-based fine-tuning configurations tailored for hate speech detection tasks, demonstrating significant computational efficiency while maintaining high performance across various model architectures.
4. Practical Contribution: We established a comprehensive evaluation framework with statistical significance testing and error analysis, specifically designed for code-mixed hate speech detection, providing benchmarks for future research in this domain.
5. Ethical Contribution: We integrated responsible AI principles throughout the model development process, including bias mitigation strategies, cultural sensitivity considerations, and ethical alignment techniques for safer digital platform deployment.

2 Literature review

Hate speech detection has become an area of considerable research interest, especially owing to widespread abusive messages in the online sphere, which also raises serious social and ethical questions. Researchers have proposed various techniques for detecting and addressing hate speech using machine learning (ML), deep learning (DL), and natural language processing (NLP). Recent progress, challenges, and future directions are discussed based on the literature from 2016 to 2024.

2.1 Traditional hate speech detection methods

Dataset Development: Early research in hate speech detection established several foundational datasets that shaped the field's development. TABHATE introduced a target-based framework for Hindi-specific hate speech analysis, addressing the need for language-specific detection systems [5]. Building on this foundation, HateXplain presented the first interpretable hate speech dataset, emphasizing both interpretability and bias analysis as critical components for responsible AI development [6]. Recently, ToxiGen contributed a large-scale machine-generated dataset specifically designed to detect adversarial and implicit hate speech, highlighting the evolving complexity of hate speech patterns and the need for more robust classification approaches [7].

Classical Machine Learning Approaches: Traditional hate speech detection methods primarily relied on conventional machine learning algorithms combined with handcrafted feature engineering. Early approaches employed Support Vector Machines (SVMs), Random Forests, and Naïve Bayes classifiers for hate speech classification tasks [8–10]. However, these classical methods have significant limitations in capturing the

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contextual nuances and semantic complexities inherent in hate speech, particularly when dealing with multilingual and code-mixed content.

Deep Learning Evolution: The emergence of deep learning architectures marked a significant advancement in hate speech detection capabilities. Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs) substantially outperform traditional machine learning approaches by better capturing sequential patterns and contextual information in text [11]. Subsequent research has demonstrated that deep learning architectures can achieve notable performance improvements over conventional methods when applied to tasks such as identifying racist and sexist content in social media posts [12].

Specialized Applications and Multilingual Approaches: Recent developments have focused on addressing specific challenges in hate speech detection across different languages and contexts. Explainable AI methods have been integrated into hate speech detection systems to enhance model transparency and responsibility [13]. Additionally, multilingual fine-tuned transformers combined with FAST-RNN architectures have shown promising results for multiclass hate speech detection in low-resource languages such as Norwegian, demonstrating the potential for cross-linguistic applications [14].

2.2 Traditional hate speech detection methods

Various datasets have been created for detecting hate speech. TABHATE is a target-based hate speech detection framework in Hindi for language-specific hate speech analysis [5]. HateXplain presented the first interpretable hate speech dataset, focusing on interpretability and bias analysis, and it is one of the first datasets in this area [6]. ToxiGen offers a large dataset generated by machines to detect adversarial and implicit hate speech, highlighting the importance of robust classifiers [7].

Classical machine learning and deep learning models have been widely used for hate speech detection. Methods such as Support Vector Machines (SVMs), random forests, and Naive Bayes have been replaced or complemented by deep learning techniques, and transformers have outperformed them in various applications, such as Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNNs) [8–10]. Recent studies examining various deep learning architectures for classifying tweets as racist, sexist, or neutral have demonstrated notable performance improvements over conventional methods for hate speech detection [11]. Emphasizing model openness and responsibility in hate speech detection, Towards Safer Online Communities uses explainable artificial intelligence (AI) methods [12]. Using FAST-RNN and multilingual fine-tuned transformers, multiclass hate speech detection in Norwegian categorizes hate speech in Norwegian [13].

2.3 Deep learning and transformer-based approaches

We organize recent transformer work into five areas: (i) supervised fine-tuning, (ii) parameter-efficient fine-tuning, (iii) prompting for low-resource scenarios, (iv) knowledge integration, and (v) domain customization. Research on hate speech detection using LLMs can be categorized into several prominent approaches, as shown in Fig. 2.

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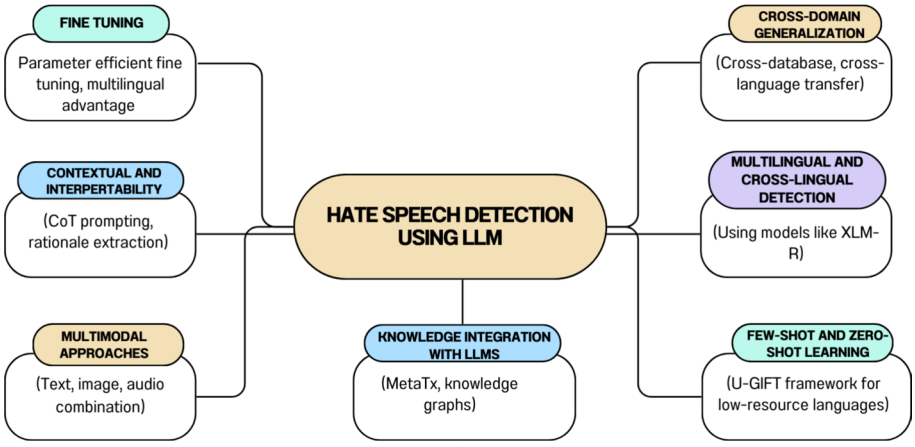

Fig. 2 Several approaches to hate speech detection using LLM

2.3.1 Fine-tuning pretrained LLMs

It has become common practice to fine-tune these pretrained LLMs on domain-specific data [14]. Sidibomma et al. (2024) created a fine-tuned model for Devanagari-scripted languages and achieved state-of-the-art results using parameter-efficient methods. These models perform well in low-resource settings and are flexible in taking advantage of multilingual and regional hate speech detection [15].

Supervised fine-tuning on task data remains strong for nuanced, implicit hate when annotated labels are available, with robust encoder baselines (e.g., RoBERTa-style and XLM-R) often ensembled or distilled to improve stability under domain shift, which is important for fair comparisons with newer decoder-style LLMs. For low-resource sequence labeling, NER-RoBERTa shows that robust RoBERTa-style encoders can be adapted effectively with careful tokenization and domain data, reinforcing the value of strong encoder baselines before moving to heavier LLMs [16].

Fine-tuning LLMs, addressing potential threats to non-white ethnic groups, and improving the accuracy of hate speech detection are crucial, especially in scenarios where high-quality annotated data are available for supervised training. This increase raised the model accuracy without damaging its recall, as its good performance was fine-tuned by the data of experienced annotators. Methodologies enabling parameter-efficient fine-tuning, such as LoRa have been used on smaller one LLMs and achieved greater than 80 percentage accuracy on the hate speech detection task [14, 15]

2.3.2 Parameter-efficient fine-tuning

Parameter-efficient fine-tuning methods have been introduced to adapt large-scale pre-trained language models effectively while minimizing computational overheads [17]. The main focus of this study was to minimize the computational costs of LLMs. Various methods, including Low-Rank Adaptation (LoRA) [18] and quantized LoRA (QLoRA) [19], have been investigated to fine-tune model parameters on hate speech datasets.

Mixed and rank-adaptive variants help stabilize training on small, shifting datasets and reduce overfitting in narrow domains, which is practical for resource-constrained moderation. We compare LoRA across diverse architectures in a Hinglish code-mixed setting and document target module choices to maintain parameter efficiency without undermining accuracy.

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2.3.3 Prompt-based, zero-shot and few-shot learning

LLMs outperform zero-shot learning in identifying hate speech without task-specific training data. This is particularly beneficial in low-resource contexts or for newly emerging types of hate speech. Recent LLMs can also work well in zero-shot learning, performing comparably to fine-tuned models [20]. Few-shot and zero-shot learning methods have been sought in the presence of small amounts of labeled data. Song et al. [21] introduced "U-GIFT", which is an uncertainty-guided model trained with few-shot learning, to mitigate hate speech in underrepresented languages. In practice, for sustained deployment and stable thresholds, fine-tuning typically remains preferable, with prompting used to bootstrap or complement the scarce labels.

2.3.4 Knowledge integration with LLMs

Knowledge integration aims to enhance the ability of large language models to detect subtle or implicit hate speech using external, structured, and contextual knowledge. Unlike few-shot learning, which lowers the dependence on large labeled datasets, knowledge integration uses tools such as knowledge graphs, hatred lexicons, or retrieval systems to augment the model's comprehension. Often overlooked by standard fine-tuning techniques, this strategy helps capture domain-specific references, cultural allusions, and historical contexts. Recent research has demonstrated how well such hybrid structures improve the performance of LLMs in several sociolinguistic settings [22]. Retrieval augmentation is especially relevant for fast-evolving euphemisms and localized slang in code-mixed communities such as the Philippines.

2.3.5 Domain-specific customization of LLMs

LLMs sometimes find it difficult to identify hate speech representations aimed at new or minority victim groups. This constraint emphasizes the need for training techniques to minimize target-group prejudice and guarantee that the models can be generalized to various forms of hate speech [23]. Hate speech manifests in various domains and requires specialized detection models. Zavira et al. (2024) described a study of LSTM-based models for hate speech detection in Telugu, which is an underrepresented language in hate speech detection [24]. Offensive Language and Hate Speech Detection in YouTube Chat references work on hate speech in live-streaming football chats, demonstrating that molecular and atomic hate speech messages can be detected using real-time detection mechanisms [25]. Recent transformer studies in Indian and other low-resource languages highlight the need for code-mixed-aware data curation, orthographic normalization, and per-domain calibration to achieve stable performance.

2.4 Interpretability and contextual understanding

2.4.1 Contextual awareness

Context and structured prompts, such as chain-of-thought (CoT) reasoning, improve model interpretability and hate speech detection performance. Such training enables models to learn the subtleties of hate speech and make accurate predictions [26, 27]. First, to overcome the black box attribute of LLMs, rationales (e.g., keywords responsible for classification) are extracted from LLMs to train interpretable classifiers, ensuring transparent and comprehensible detection [5]. LLMs are effective at capturing the complexities of context, which is crucial for the precise identification of hate speech.

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Using carefully designed reasoning prompts, LLMs can leverage the knowledge acquired to achieve superior performance in detecting hate speech [7]. Rationale distillation and explanation-enhanced datasets offer a path to policy transparency without sacrificing accuracy.

2.4.2 Explainability and transparency

CoT prompts and post-hoc explanations are differentiating factors that make LLMs more interpretable. This not only enhances the accuracy of the model but also produces the rationale behind the decision-making process, resulting in more interpretable and trustworthy models [28]. Although LLMs have some benefits, they can make mistakes. Misclassifications can occur because some requests would take advantage of model flaws. These "jailbreak" prompts require strong defenses to ensure consistent behavior [27]. This study introduced a method for improving and simplifying hate speech detection better and simpler while preserving the model's ability to detect these messages [24]. Operationally, high-stakes or borderline cases (e.g., satire, reclaimed slurs, and sarcasm) still benefit from human escalation.

2.5 Multilingual, cross-lingual, and code-mixed capabilities

2.5.1 Multilingual generalization

Hate speech surpasses linguistic barriers; therefore, multilingual solutions are required to address it in the future. In the cross-lingual setting, XLM-Roberta and other similar models performed excellently. Using XLM-Roberta, Hashmi et al. (2024) designed a hate speech index based on attention-assisted LSTMs, suggesting that it can be generalized well across languages [29]. LLMs generalize well to multiple languages, including low-resource languages such as Bengali and Hindi. For example, GPT−3.5 Turbo has remarkably good precision in interpreting hate speech in Bengali by a large margin compared with traditional approaches [28].

Hate speech is multilingual. Models such as XLM-RoBERTa, along with newer architectures, have demonstrated robust cross-lingual performances. Emerging research has highlighted the need for few-shot and zero-shot learning strategies (Song et al., 2025) to address data scarcity in underrepresented languages [21].When computing is constrained, translation-based pipelines remain viable; cross-lingual transfer is feasible but requires calibration and bias audits to avoid disparate error rates.

2.5.2 Code-mixed language challenges

Detecting hate speech in code-mixed languages presents unique challenges owing to the blending of linguistic structures and vocabulary. Prathap et al. [30] proposed a deep-learning approach for detecting hate speech in Tamil code-mixed languages, highlighting the potential of neural networks for handling complex linguistic patterns. Similarly, Sreelakshmi et al. [2] discussed the challenges of detecting hate speech in code-mixed languages, emphasizing the need for robust models that can handle this linguistic diversity.

Hidayatullah et al. [31] provided a comprehensive review of hate speech detection in code-mixed languages, covering various approaches and techniques. Their work underscores the importance of developing models that can effectively process and understand code-mixed text. Anbukkarasi et al. [32] explored the use of deep learning for hate

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speech detection in code-mixed text, demonstrating the efficacy of such approaches in improving detection accuracy. Chanda et al. [33] focused on detecting hate speech in code-mixed social media texts, highlighting the challenges posed by informal content. Their research contributes to the development of more effective hate speech detection systems for real-world applications. Table 1 shows the well-structured work performed on code-mixed Hindi–English Texts using traditional and recently developed LLM-based approaches. Orthographic inconsistencies in related scripts (e.g., Arabic-script variants) degrade the embeddings. Consistent normalization improves robustness, which is a transferable insight for code-mixed processing [34].

Table 1 Comparison of models and methodologies for hate speech detection in Hindi-English code-mixed data

References	Title	LLMs/model	Datasets	Key findings	Methodology
Mazumder et al. [35]	Improving Code-Mixed Hate Detection by Native Sample Mixing: A Case Study for Hindi-English Code-Mixed Scenario	mBERT, XLM, XLM-R; SVM, RF, NB; Enhanced MLMs with extra transformer layers	112,718 tweets; 4575 manually annotated code-mixed tweets	Native data improves MLM performance and focus on hate words	Native + code-mixed dataset constructed, traditional and MLM models evaluated, and attention scores visualized to assess multilingual model performance
Sreelakshmi et al. [2]	Detection of Hate Speech Text in Hindi-English Code-mixed Data	SVM (RBF, Linear), Random Forest	10,000 Hindi-English code-mixed samples	fastText + SVM-RBF best performance; char-level features most effective	Used fastText embeddings; trained and compared ML classifiers
Thomas et al. [36]	A Dataset of Hindi-English Code-Mixed Social Media Text for Hate Speech Detection	SVM (RBF), Random Forest	4,575 Hindi-English tweets (from 112,718 collected)	Char n-grams with SVM gave 71.7% accuracy	Extracted features like n-grams, lexicons; chi-square feature selection; SVM + RF evaluation
Kumar et al. [4]	Multi-task Detection of Harmful Content in Code-Mixed Meme Captions Using LLMs	BART MNLI, Flan-T5 XXL/Base, Mistral 7B, Tiny Mistral	Multi-Bully Dataset (code-mixed meme captions)	Multi-task classification improves detection performance	Zero-/few-shot prompting, LoRA tuning; multi-task labels: cyberbullying, sentiment, emotion, etc
Vashistha and Zubiaga [37]	Online Multilingual Hate Speech Detection: Experimenting with Hindi and English Social Media	CNN-LSTM, BERT variants	Aggregated Hindi, English, Hinglish datasets	Multilingual models outperform monolingual ones	Dataset fusion; online feedback-based model retraining
Sharma et al. [38]	THAR - Targeted Hate Speech Against Religion	MuRIL, Transformer-based, DL models	THAR: 11,549 code-mixed comments	F1 score of 0.78 on hate detection tasks	Annotated by 5 users; Fleiss' Kappa used; tested DL/Transformer models
Yadav et al. [39]	Hate Speech is not Free Speech: Explainable ML for Hate Speech in Code-Mixed Languages	Bernoulli Naive Bayes (67.61% accuracy)	HASOC ICHCL	Slur bias affects results; LIME improves explainability	CountVec/TF-IDF + Naive Bayes; interpretability via LIME
Gupta et al. [40]	Hindi-English Code Mixed Hate Speech Detection using Char-Level Embeddings	GRU with Attention	Hinglish dataset	GRU-Attn with char embeddings outperforms others	Char embeddings + hybrid deep learning classifier (GRU + Attention)

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2.6 Multimodal and cross-domain approaches

2.6.1 Text and image / video models

To address the multifaceted issue of online hate speech, multimodal models that use combinations of textual, visual and auditory data have been proposed. These models, such as fine-tuned Llama 2, achieve high performance in hate speech detection across modalities [6, 10]. However, cross-domain generalizability is an essential issue in hate speech detection, and research has revealed that LLMs can achieve acceptable results in terms of generalization across different databases (e.g., hate speech annotated tweets) and even different languages, thereby greatly expanding their use in diverse settings [26, 41]. Recent transformer attention frameworks for memes have reported gains via scalable multi-level attention and graph/context fusion, underscoring the value of conversational and visual cues. Cross-modality transfer with in-context learning has been explored to bridge text-only training and meme-like inferences. LLaMA-Adapter+ with meta-reasoning prompting (MRP) integrates task-adaptive reasoning with lightweight adapters for efficient multi-modal modeling, highlighting a practical path to fuse text–vision signals without full retraining [42].

Despite the advancements in LLMs, challenges such as ethical deployment and model bias still need to be addressed in future studies. High-quality data and responsible AI implementation practices are essential for overcoming these difficulties and improving the robustness of LLM-based hate speech identification systems [37, 38].

Multimodal models that extend beyond text to incorporate images, videos, and text have recently attracted attention. Hee et al. (2024) present a comprehensive picture of what is going on in the field of hate speech moderation with specific reference to the increasing importance of multimodal approaches and the transformative power of large language models [7].

The increasing amount of hateful content in the form of images has resulted in the use of multimodal approaches (a combination of text and images). Contrastive Language-Image Pre-Training (CLIP)-based detection of hate speech in members uses CLIP to classify hate speech in memes with high accuracy [39].

2.6.2 Real-world application domains

Multimodal hate speech moderation: The combination of LLMs with multimodal models has the potential to identify and moderate hate speech on social media platforms, where multimedia content is common. This is a particularly elegant way to address the intricate relationships between textual, visual, and auditory signals [10]. Countering Hate Speech in Political Discourse: LLM models have been successfully used to detect and counter hate speech in the context of political discourse, as reflected in the 2022 Russia-Ukraine war communication. Using retrieval-augmented generation and instruction-based models, successful workflows have been designed to counter, diffuse, and classify hate [40]. Supporting Human Raters: LLMs have been used to aid humans in identifying harmful content such as hate speech. The workload of human review problems is reduced by pre-filtering, error detection, and LLM retrieval [43]. Operationally, pipelines benefit from periodic refresh with hard negatives and the use of thread-level context and author history features to improve cross-domain generalization. The human-in-the-loop approach remains critical for ambiguous or high-impact actions.

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2.7 Challenges and limitations

2.7.1 Data imbalance and bias

LLMs are also trained on imbalanced datasets and may inherit biases, resulting in oversampling or undersampling of some categories of hate speech. Addressing this issue will involve vigilance in training data curation and post-hoc strategies for dealing with bias.

2.7.2 Ambiguity in hate speech

The subjective nature of hate speech makes it difficult to annotate and classify. For example, models may find it difficult to identify subtle hate speech, sarcasm, or reappropriated slurs, as mentioned in [40] when studying Italian-language slurs.

2.7.3 Generalization and cultural bias

Different languages and cultures may present varied effectiveness in the LLM. For example, although English models tend to be quite good at the task, Hinglish is a special problem that requires tailored solutions [44]. Hate speech detection models suffer from bias, which is a critical issue in this field. Other studies have demonstrated the mislabelling that annotators tend to make, particularly owing to their dialectal bias [6], and the disproportionate flags of AAE are toxic. The risk of racial biases in hate speech detection [36] explored annotation bias in popular datasets and proposed dialect-priming mechanisms to remove biases [41].

2.7.4 Interpretability, errors, and model vulnerabilities

LLMs have proven to be effective and powerful tools. However, they are black boxes in terms of their interpretability. For example (Fuseini and Mumuni) introduce their work in (2025) introduced this problem to provide human-understandable reasons for model predictions [44]. Although LLMs have some benefits, they can make mistakes. Misclassifications could occur because some requests would take advantage of model flaws. These "jailbreak" prompts require strong defenses to ensure consistent behavior [45]. Figure 3 illustrates the key challenges and recent methodological advancements in LLM-based

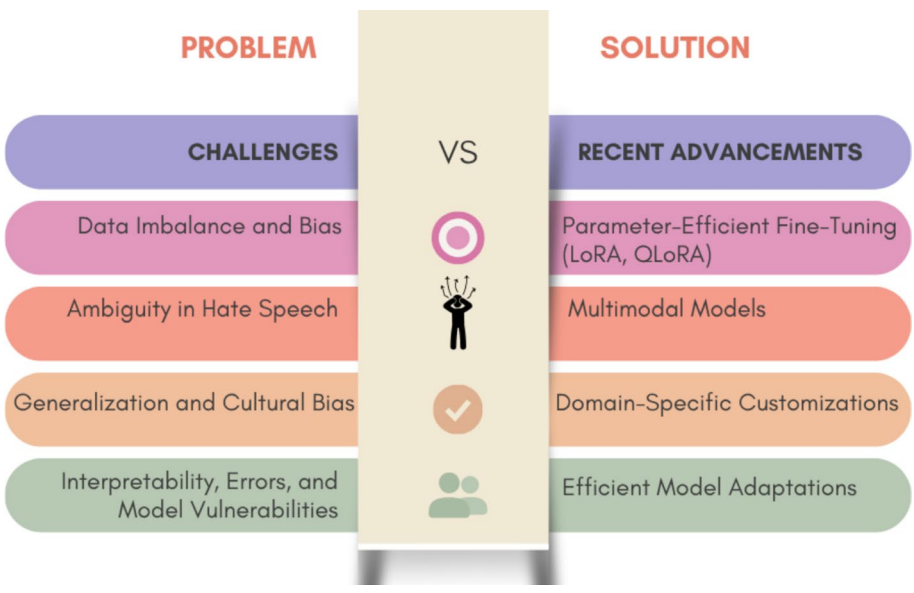

Fig. 3 Challenges and recent advancements in LLM-based hate speech detection

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hate speech detection, providing a structured overview of the current limitations and emerging solutions.

Section 3.6 includes a thorough investigation of these limitations and challenges, as well as our suggested solutions to address them.

2.8 Best practices for using LLMs in hate speech detection

2.8.1 High-quality datasets

To perform as well as possible, the training data should be of high quality and diverse. For example, annotating research assistants/crowdsourced data across ethnic groups can lead to more accurate LLM [46].

2.8.2 Prompt design and fine-tuning

Prompt engineering is a key factor in the performance of LLMs. Contextual and explanatory informative prompts can improve both accuracy and interpretability [47].

2.8.3 Human-in-the-loop moderation

Although high LLM accuracy can be achieved, human supervision is crucial to cover edge cases and ensure ethical aspects. Combining LLM with human raters may enhance the detection process and increase reliability [48].

2.8.4 Model updating and lifecycle management

Owing to the time-varying nature of hate speech, the periodic updating and optimization of LLMs are required to maintain their efficacy. This includes the detection of new types of hate speech and changes in linguistic styles [6]. In deployment, model cards, bias audits, and periodic refreshes with hard negatives improve trust and robustness of the model.

2.9 Summary of research gaps

The complexity of languages, differences in grammar, variations in transcription, and social factors make it challenging to recognize hate speech in code-mixed languages such as Hinglish (Hindi-English). Code mixing creates non-standard vocabulary, casual syntax, and phonetic spelling, which disturb traditional NLP pipelines and can potentially compromise the performance of pretrained language models (LLMs).

Moreover, few studies have methodically examined the effects of parameter-efficient fine-tuning techniques, such as Adapters and LoRA, particularly in code-mixed languages. This remains a major research gap because such methods can fit strong models to resource-constrained multilingual situations with lower computing costs. Future research should focus on creating useful and complete code-mixed collections to address these limitations. They should also conduct research on domain adaptation and measure the outcomes of using lightweight fine-tuning across different languages. We explicitly compare our model with strong transformer families (RoBERTa/XLM-R encoders, LLaMA-adapter variants, compact "tiny LLMs") and explain why our setting differs (Hinglish code-mix focus, transparent PEFT configurations with architecture-specific target modules, and a responsibility-oriented evaluation protocol).

Figure 4 shows a futuristic roadmap for detecting hate speech using LLM. It presents a concise roadmap for advancing LLM-based hate speech detection across four pillars:

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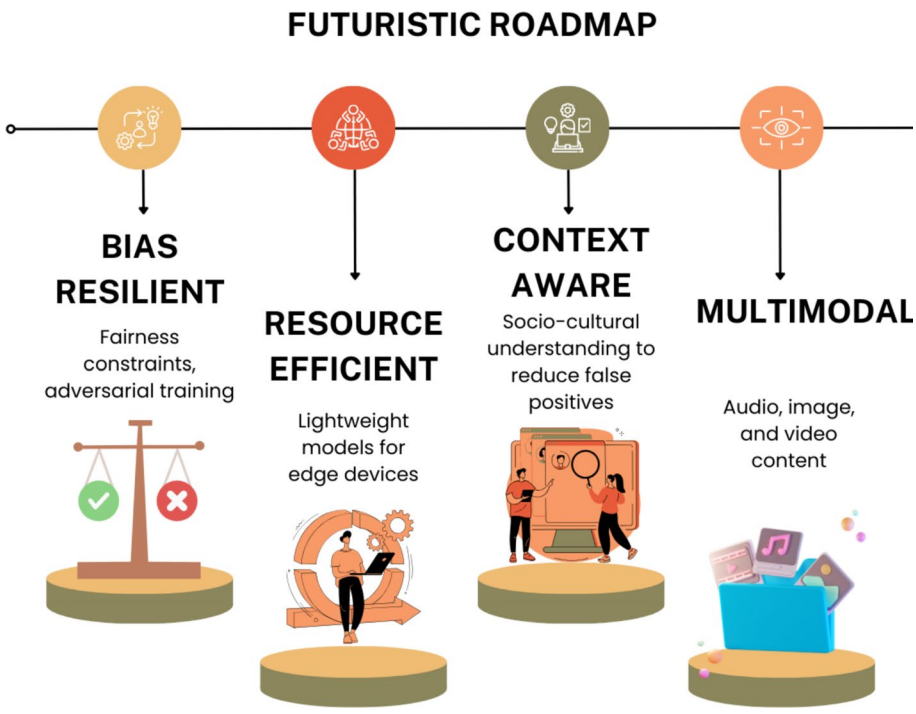

Fig. 4 Envisioning the future of LLM-based hate speech detection

bias-resilient models that apply fairness constraints, auditing, and adversarial training to reduce disparate error rates; resource-efficient adaptation that leverages parameter-efficient fine-tuning (e.g., LoRA/adapters) and lightweight inference for constrained or edge deployments; context-aware detection that incorporates socio-cultural cues in English–Hinglish code mix to lower false positives and improve robustness; and multimodal moderation that extends beyond text to audio, image, and video for cross-platform reliability. Together, these pillars translate the study's findings into a practical path from current experiments to scalable, safe deployment, with details aligned with Sects. 3, 4 and 5 for methods, evaluation, and future directions.

3 Methodology

3.1 Dataset

The Reddit text dataset was selected as the primary source because its India-focused communities naturally produce authentic English–Hinglish code-mixed text with informal orthography, mixed scripts, and slang that mirror real moderation challenges better than clean, monolingual corpora (e.g., tweets with heavy filtering). The platform's threaded discussions provide a richer context and topic diversity (politics, religion, sports, daily life), enabling models to learn both explicit and implicit hate patterns and avoid overfitting to a single genre or hashtag style. To ensure ethical and legal use, only publicly available posts were collected via the official API, user identifiers were removed, and sensitive tokens were masked in the exemplars. Table 2 shows a sample dataset of Reddit Hinglish and English social media posts in text format because the sensitive post "*" is marked in some words. In the Reddit dataset, the classes were labeled 0 and 1, where 0 represented no hate speech and 1 represented hate speech.

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Table 2 Classification of reddit social media posts

Post	Class
anup bhai mast kaam karta h	0
jab bestfriend ho to cham**r gf ki kya jarurat	1
at sufi meet modi lauds islam as a religion of peace	0
yahan to alag hi checking chal rahi hai vro	0
waise wo hai chu***a usme koi shaq nhi	1
modi hai toh mumkin hai	0
ground report behind Karnataka bjps ji**di murder list more than hindutva and islamist clashes	1
kya matlab karachi mei bhi diwali manate hai	0
maan i hate these jih*di fuc***rs by heart	1
witnessing pollution of delhi from air	0

Fig. 5 Word cloud of informative tokens in Hinglish and English code-mixed reddit hate speech data

We constructed a custom Reddit dataset to capture real-world English–Hinglish code mix and context diversity that are underrepresented in prior corpora, enabling the rigorous evaluation of parameter-efficient LLM fine-tuning for hate speech detection.

Figure 5 presents a word cloud of informative tokens in Hinglish and English code-mixed Reddit hate speech data, which provides the most frequent and contextually significant terms used in the dataset that lead to the recognition of hate speech patterns in both languages.

3.2 Data preprocessing

The data preprocessing steps included tokenization, normalization, and bias mitigation strategies.

Preprocessing pipeline. To prepare English–Hinglish posts for modeling, we implemented a four-stage pipeline tailored to the code-mixed text.

Stage 1: Text normalization. We lowercase Latin tokens, preserve Devanagari script, and normalize repeated characters (e.g., 'goooood'→'good') while retaining emphasis markers when they are semantically informative. URLs, emails, and user mentions were replaced with the tokens [URL], [EMAIL], and [USER], respectively. Emojis were mapped to polarity tags when available, to preserve sentiment cues. Contractions and common Hinglish spellings were standardized using a custom lexicon (e.g., 'ppl'→'people', 'tumhara' variants unified), and stray punctuation/HTML was removed.

Stage 2: Tokenization and language ID. Posts were segmented using a rule-based tokenizer augmented with a lightweight LID module that tagged each token as En/Hi/Other. Mixed-script Hindi in Latin characters (e.g., 'yaar', 'kitna') was detected via a curated Hinglish lexicon and syllable patterns. This step yields En/Hi tags per token and enables code-switching analysis and vocabulary filtering.

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Stage 3: Noise reduction and bias control were performed. We remove boilerplate (quotes, signatures), collapse near-duplicates using MinHash on character trigrams, and filter posts that are shorter than five tokens. To prevent slur-shortcut learning, we balanced the frequency of high-risk keywords across classes and masked rare identity terms when they were not the target of hate to reduce spurious correlations. Class balance is rechecked after filtering.

Stage 4: Final checks and serialization of the data. We validated the language-mix ratio for posts labeled as code–mixed, requiring at least ≥ 10 percentage of tokens from each language to ensure sufficient signal for code–switch modeling. When a post is a direct reply and its interpretation depends on the parent, we retain the minimal necessary context to preserve semantics. The cleaned instances were then serialized to JSONL with the following fields: {id, text, label, lang_tags, meta}. Finally, we created stratified *train/test* splits by both class label and subreddit to maintain class balance and topic diversity across splits.

Input: Dataset of Reddit posts D, Trained BoW Classifier C, Hate keywords H, Threshold $\theta = 6$
Output: Annotated dataset with final labels L

1: Initialize annotated dataset $L \leftarrow \emptyset$
2: **for** each post $p \in D$ **do**
3: Preprocess p (tokenize, clean, lowercase, etc.)
4: Use classifier C to predict label ML_label for p
5: **if** ML_label is high confidence (probability $\geq \theta/10$) **then**
6: Assign final label $label_f \leftarrow ML_label$
7: Add $(p, label_f, \text{“ML auto-label”})$ to L
8: **else**
9: Annotator1 scores p with $score_1 \in [0, 10]$
10: Annotator2 scores p with $score_2 \in [0, 10]$
11: $label_1 \leftarrow (score_1 \geq \theta)$
12: $label_2 \leftarrow (score_2 \geq \theta)$
13: **if** $label_1 == label_2$ **then**
14: $label_f \leftarrow label_1$
15: **else**
16: Third reviewer scores p with $score_3 \in [0, 10]$
17: $label_f \leftarrow (score_3 \geq \theta)$
18: **end if**
19: Add $(p, label_f, \text{“Human adjudicated”})$ to L
20: **end if**
21: **end for**
22: **return** L

Algorithm 1 Annotation Algorithm for Reddit Posts

In Algorithm 1, the annotation workflow combines a fast machine screen with a careful human review to produce reliable labels for Reddit posts in English–Hinglish. First, each post was lightly cleaned and tokenized so that the same text was seen by both machine and human annotators (lines 2–3). A trained baseline classifier C then assigns a probability that the post is hate speech (line 4). If this probability is high enough at least $\theta/10$, which equals 0.6 when $\theta = 6$ the machine label is accepted directly to reduce human effort on obvious cases (lines 5–7).

Otherwise, the post is routed to two independent annotators, who each assign a score from 0 to 10 using a short rubric (explicit slurs, target group, dehumanization, threats,

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coded expressions) (lines 9–10). Each score is converted to a binary label by comparing it to the decision threshold θ: scores $\geq \theta$ map to hate (1), and scores $< \theta$ map to non-hate (0) (lines 11–12). If the two labels match, the label becomes final (lines 13–14). If they disagreed, a third senior reviewer inspected the post (and, when needed, its parent comment for context) and gave a final score; this was converted with the same threshold rule to obtain the definitive label (lines 15–17).

Every record is saved together with its provenance tag, either ML auto-label or human adjudicated, so that later analyses can trace how the label was produced (lines 7, 19–22).

3.2.1 Toxicity filtering and ethics checks

This study applies a conservative safety screen on the training split only to reduce annotator exposure and discourage shortcut learning while avoiding any leakage into evaluation. After normalization (Sect. 3.2), each post receives a toxicity probability from a third-party toxicity classifier; the score and resulting action are recorded as `tox_score` and `tox_action` $\in \{\text{pass}, \text{review}, \text{mask}, \text{exclude}\}$. Items with toxicity scores ≥ 0.85 are routed for human review prior to training set inclusion; items with scores ≥ 0.98 are masked or excluded if they contain graphic threats or highly explicit content. Neutral sentences containing identity terms were retained and never filtered solely on the presence of identity tokens. Filtering decisions require contextual cues (e.g., threats, dehumanization) and human confirmation in borderline cases. No toxicity filtering has ever been applied to the held-out test set; only minimal masking of overt slurs is applied to exemplars in figures/tables for reader safety. This placement after normalization and before class balancing stabilizes the classifier behavior, preserves class distributions, and prevents biasing the evaluation.

3.2.2 Annotation validation and agreement

3.2.3 Annotation protocol

Two independent annotators labeled all items using a concise rubric focused on explicit slurs, targeted groups/individuals, dehumanization, threats, and coded derogation. Minimal parent context was provided when a post's meaning depended on the reply chain. Disagreements were adjudicated by a senior reviewer. Prior to full labeling, the annotators completed a calibration batch with feedback, and weekly spot checks sampled a small percentage of new labels for quality assurance.

3.2.4 Overlap and metrics

Inter-rater reliability was computed on the human-labeled overlap only (machine pre-screened labels were not included). Pairwise agreement (Cohen's χ) shown in Table 3 Annotator 1 vs Annotator 2: n=2335, raw agreement=0.862, χ=0.724; Annotator 1 vs

Table 3 Inter-annotator agreement on the Hinglish dataset

Pair	*n*	Raw agreement	κ
Annotator 1 vs. Annotator 2	2335	0.862	0.724
Annotator 1 vs. Annotator 3	2344	0.902	0.803
Annotator 2 vs. Annotator 3	2341	0.876	0.750
Fleiss (3 annotators)	2328	–	0.759

Pairwise rows report raw agreement and Cohen's χ; the last row reports Fleiss' χ on items labeled by all three annotators

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Annotator 3: n=2344, raw agreement=0.902, χ=0.803; Annotator 2 vs Annotator 3: n=2341, raw agreement=0.876, χ=0.751. On the triple-labeled subset (n=2328), Fleiss' χ=0.759. These values indicate substantial agreement for the binary hate/non-hate task. The individual annotator accuracy versus the final adjudicated label was 0.938 (Annotator 1), 0.916 (Annotator 2), and 0.951 (Annotator 3), and the overall adjudication rate was modest, reflecting a stable consensus under the rubric.

3.3 Model architecture and fine-tuning

Our experiments used several state-of-the-art LLMs.

- DeepSeek R1
- LLaMA 3
- DistilBERT
- Mistral
- Phi 3.5
- Gemma 3
- ModernBert

We performed fine-tuning primarily using parameter-efficient approaches.

- Adapters
- Low-Rank Adaptation (LoRA)

We fine-tuned the pretrained models using LoRA adapters, which allowed efficient adaptation by injecting low-rank trainable matrices into the transformer attention layers.

3.4 Experimental setup and hyperparameter tuning

Each model was fine-tuned using a custom Hinglish hate speech dataset with an 80/20 train-test split. Training was performed on NVIDIA A100, Tesla T4, and RTX 3060 GPUs, depending on resource availability, as detailed in Table 5. To ensure fairness, all the models were trained for five epochs under consistent experimental conditions. A grid search was applied for learning rates ($1e^{-4}$–$2e^{-5}$) and batch sizes (8, 16, and 32) adjusted according to the model size. The model performance was evaluated using the following standard classification metrics: **precision**, **recall**, **F1-score**, and **accuracy**. For parameter-efficient tuning, *Low-Rank Adaptation (LoRA)* was employed across models with task-specific modifications in rank (r), alpha (α), and dropout rate. The LoRA configurations for each model, including the targeted modules for injection, are listed in Table 6.

3.5 Evaluation metrics and analysis

To evaluate the model, we measured its accuracy, precision, recall, and receiver operating characteristic (ROC) curves. For training, we examined the adapter settings, LoRA configuration, number of epochs, and training loss.

Statistical significance testing, error testing, and interpretability analyses were performed to analyze the results of the study. We used more in-depth metrics, such as ROC-AUC and confusion matrices, for an in-depth evaluation. Figure 7 shows the adapter architecture, which is a lightweight and efficient method for fine-tuning the pretrained models. It adds small trainable modules (adapters) to the existing layers of

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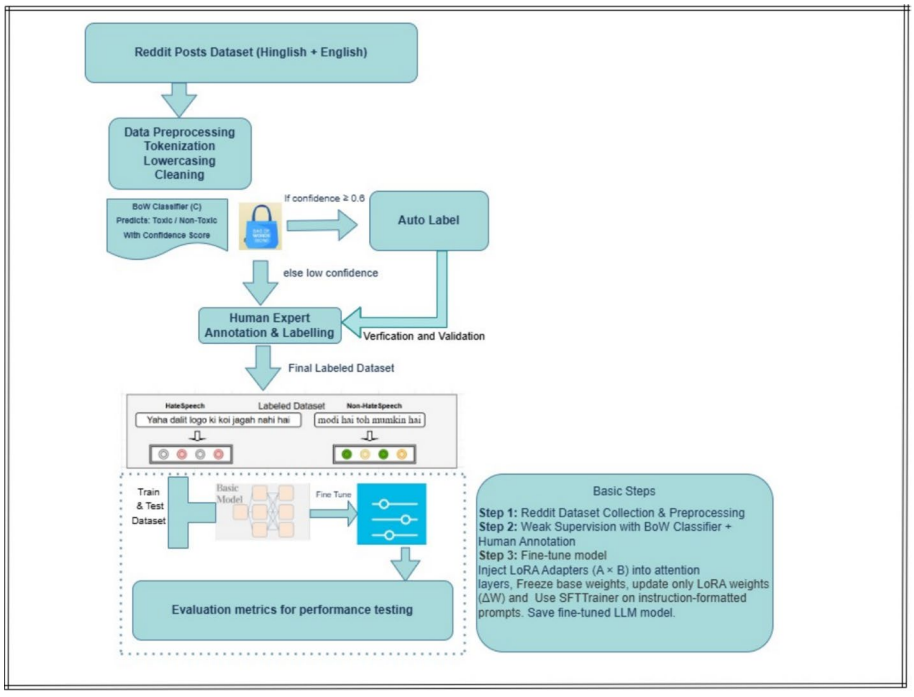

Fig. 6 Methodology workflow: hybrid annotation and fine-tuning framework for hate speech detection in a custom reddit dataset

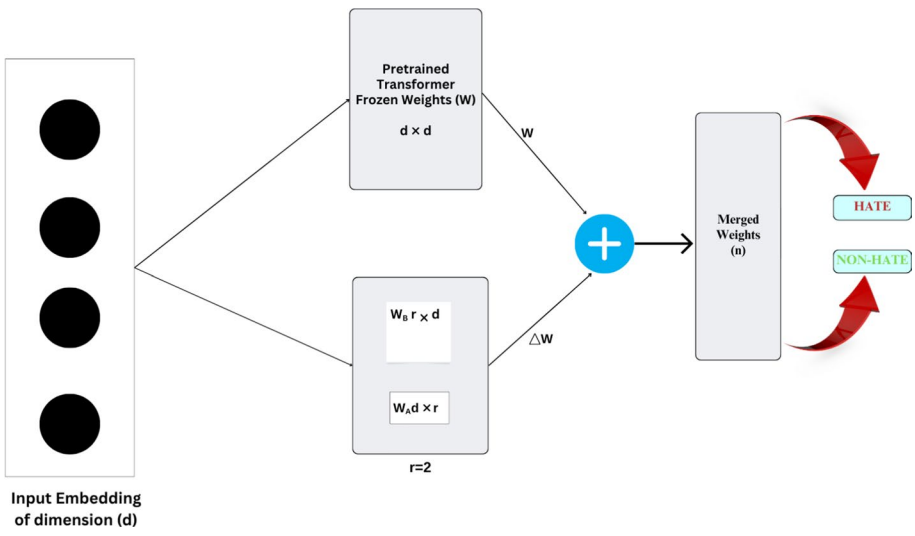

Fig. 7 Adapter architecture

the model, allowing task-specific modifications, whereas most pre-trained parameters remain frozen.

Figure 6 shows the end-to-end workflow of the proposed hate speech detection model, which is based on the Hinglish Reddit dataset. The pipeline involves weak supervision through a classifier with a BoW classifier integrated with human expert annotations and fine-tuning with a LoRa-based version of the LLM for better classification accuracy (Fig. 7).

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Table 4 Evaluation results of fine-tuned transformer models on the custom dataset

Model	Precision	Recall	F1-score	Accuracy
DeepSeek R1	0.82	0.79	0.81	0.79
Gemma 3	0.82	0.78	0.80	0.78
Mistral	0.82	0.78	0.80	0.78
Phi 3.5	0.81	0.78	0.80	0.77
LLaMA 3	0.78	0.73	0.75	0.73
ModernBERT	0.74	0.72	0.73	0.72
DistilBERT	0.68	0.68	0.68	0.68

Table 5 Comparison of fine-tuning time of different transformer-based models

Model	Fine-tuning epochs	Platform (GPU)	Fine-tuning time
Gemma-3 4b-it	5	Tesla T4 14.741 GB	81 mins
Distilbert-base-uncased	5	Tesla T4 14.741 GB	49 mins
Deep Seek R1	5	Nvidia RTX3060 (12GB)	52 mins
Llama 3	5	NVIDIA A100 40GB	51 mins
Modern BERT	5	Nvidia A100 GPU 40 GB	6 mins
mistral-7b-bnb-4bit	5	Nvidia RTX3060 (12GB)	87 mins
Phi 3.5	5	Nvidia RTX3060 (12GB)	70 mins

In this study, we apply *low-rank adaptation* (LoRA) to fine-tune the LLM models. Instead of updating all the weights of the transformer, the LoRA introduces trainable rank-decomposed matrices A and B into the attention layers, allowing efficient training with significantly fewer parameters. Specifically, the LoRA modifies the linear projections q_{proj} and v_{proj} as follows:

$$W_{adapted} = W + \Delta W = W + BA \tag{1}$$

3.6 Summary and our approach to challenges

Our proposed approach addresses the limitations discussed in Sect. 2.6. To deal with data imbalance and bias (2.6.1), we created a high-quality, balanced Hinglish hate speech dataset based on Reddit sources and expert validation. Human-in-the-loop filtering was used to handle ambiguity in hate speech (2.6.2), which included sarcasm and subtle slurs. To eliminate cultural bias and increase generalization (2.6.3), we implemented Code-mixed data augmentation based on South Asian language settings. Finally, to improve interpretability and minimize compute cost (2.6.4), we used LoRA-based fine- tuning for domain-specific adaptation of large models while retaining efficiency and ethical alignment

4 Results

In this section, we present the detailed results, including the accuracy, precision, recall, and ROC curve, as well as the specific configurations for adapters, LoRa, epochs, steps, and training time. The final evaluation compared the performances of DeepSeek R1, Llama 3, DistilBERT, ModernBERT, Mistral, and Phi 3.5 on hate speech detection tasks using a custom dataset.

The comparison metrics of the transformer-based models (precision, recall, F1 score, and accuracy) for hate speech detection are listed in Table 4 for all the fine-tuned models. DeepSeek R1 had the highest performance (0.79 for both the F1 score and accuracy), which we would expect from DeepSeek R1, given its ability to model complex

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and contextual inputs. Mistral and Gemma 3, with F1 scores of 0.78, were maintained as good in every run, indicating a reliable classification. Although LLaMA 3 achieved a recall of (0.73 macro avg), the precision was (0.78 macro avg), and the overall F1 score was (0.72), indicating a bias towards false positives. Importantly, all metrics, particularly the F1 score, showed balanced but low results for DistilBERT and ModernBERT, which is consistent with their lighter architecture. These results reinforce the notion that the complex semantics of hate speech, as captured by modern LLMs such as DeepSeek R1, Gemma 3, and Mistral, provide a meaningful advantage, particularly in code-mixed text scenarios. We report the precision, recall, F1, and accuracy on the held-out test set. Uncertainty was estimated via nonparametric bootstrap over test examples (1000 resamples, 95 percentage CIs). Pairwise differences in F1 between the models were considered significant if the 95 percentage CI of the difference excluded zero. All intervals were computed from fixed test predictions with no retraining.

The model fine-tuning configuration and time required for training are presented in Table 5, which shows the computational intensity of some modern architectures based on the transformer. The fine-tuning time spanned a wide range, although most models were trained for five epochs on comparable hardware platforms (mostly Nvidia RTX3060 with 12GB of memory and an A100 GPU). Deep Seek R1 ran on an Nvidia RTX3060 (12GB) GPU and required 52 min of training time, and ModernBERT had the shortest training runtime (6.25 min) but also performed the average in this short duration compared to other LLM models. In contrast, for context, the training of DistilBERT and Gemma-3 4b-it was achieved in less than an hour (49 and 81 min, respectively), emphasizing the efficiency and deployment potential of these models in resource-limited settings. The trade-off between performance and efficiency in training is clear, whereas Mistral and Phi3.5 provide good accuracy with moderate training times. This

Table 6 Hyperparameters for different transformer-based models

Hyperparameters	Distil BERT	Mistral	Modern BERT	Phi 3.5	Gemma 3	Llama 3	DeepSeek R1
Epoch	5	5	5	5	5	5	5
Target Modules	["q_lin", "v_lin", "k_lin", "out_lin", "ffn.lin1", "ffn.lin2"]	["q_proj", "k_proj", "v_proj", "o_proj", "gate_proj", "up_proj", "down_proj"]	["Wqkv", "out_proj", "Wi", "Wo"]	["q_proj", "k_proj", "v_proj", "o_proj", "gate_proj", "up_proj", "down_proj"]	["q_proj", "k_proj", "v_proj", "o_proj", "gate_proj", "up_proj", "down_proj"]	["q_proj", "k_proj", "v_proj", "o_proj", "gate_proj", "up_proj", "down_proj"]	["q_proj", "k_proj", "v_proj", "o_proj", "gate_proj", "up_proj", "down_proj"]
Trainable Parameters	2.786%	0.015%	15.30%	0.015%	0.82%	0.52%	0.52
LoRA_alpha	32	32	258	32	32	16	32
Rank (r)	16	16	128	16	16	16	16
LoRA_dropout	0.01	0.05	0.01	0.05	0.05	0	0.05
Batch Size	32	16	16	8	16	16	16

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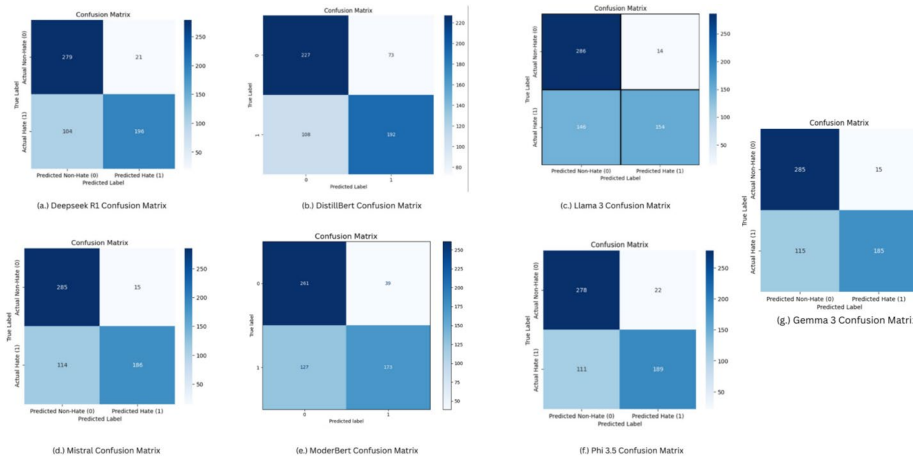

Fig. 8 Confusion matrix for classifiers

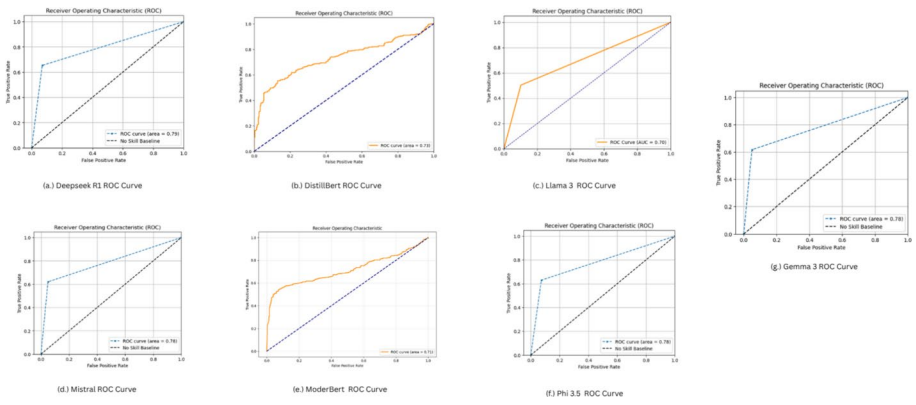

Fig. 9 ROC curve for classifiers

demonstrates that parameter-efficient models can be scaled for collection tasks without a significant performance penalty.

Table 6 lists the hyperparameter set for multiple transformer-based models, covering the main settings, such as the target modules, batch size, number of epochs, rank, LoRa dropout, LoRa alpha, and the training parameters. Each model configuration was created to achieve the best results for the hate speech detection task.

The confusion matrices for the six large language models (LLMs) on hate speech detection are shown in Fig. 8: (a) DeepSeek R1, (b) DistilBERT, (c) Llama 3, (d) Mistral, (e) ModernBERT, and (f) Phi 3.5. Each matrix shows the distribution across the predicted and actual labels for two classes: non-hate (label 0) and hate (label 1).

The performance of the DeepSeek R1 model was much more balanced, with 279 true negatives and 196 true positives, which is an improvement, although it incorrectly classified 104 hate speech instances as non-hate speech. DistilBERT operates conservatively, resulting in fewer false positives but a higher false negative count (108) and only 73 true positives. LLaMA 3 performed well overall, with 286 true negatives and 154 true positives; however, 146 false negatives and Gemma 3 showed reliable performance, with 285 true negatives, 185 true positives, and 115 false negatives, indicating a stronger capability to correctly detect hate speech.

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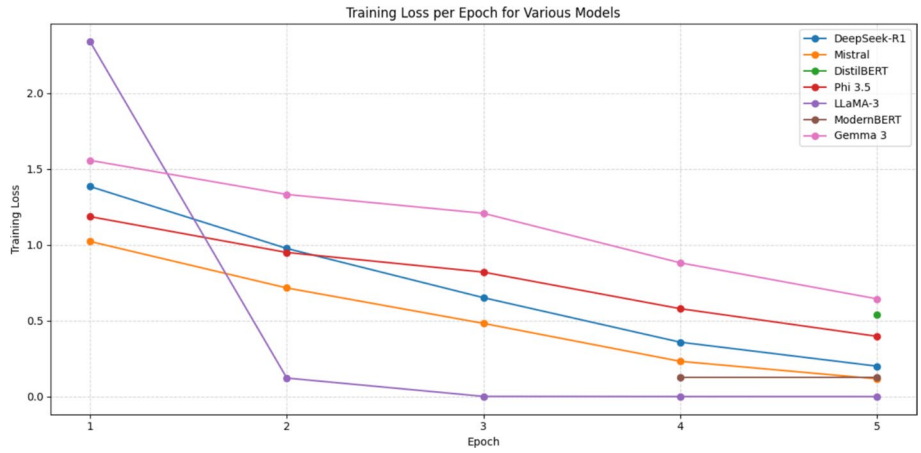

Fig. 10 Training loss for various language models on hate speech detection task

Mistral had the best precision-recall balance, delivering 285 true negatives and 186 true positives with only 15 false positives. The performance was on par with that of ModerBERT and Phi 3.5, with ModerBERT having slightly more false negatives (127) and Phi 3.5 indicating a more uniform overall distribution between false positives and false negatives.

Collectively, these confusion matrices highlight the identification of hate speech that each of the models excels at or lacks, with DeepSeek R1, Gemma 3, Mistral, and LLaMA 3 showing a stronger generalization over both classes.

Figure 9 shows the comparison of six language models, including DeepSeek R1, Gemma 3, DistilBERT, LLaMA 3, Mistral, ModernBERT, and Phi−3.5, in terms of the performance of the language model that can be performed by the ROC curves of those language models on the classification task. The ROC curve shows the ability of each model to differentiate between classes, and the area under the curve (AUC) was the main criterion for the comparison. DeepSeek R1 produced the highest AUC (0.79), followed closely by Mistral, Gemma 3 and Phi−3.5 (0.78). With an AUC of 0.73, DistilBERT performed modestly, followed by Llama 3 (AUC = 0.70), and ModernBERT (AUC = 0.71). Taken together, these findings imply that Deepseek R1, Mistral, Gemma 3, and Phi−3.5, in this study for binary classification, outperform other models because of their higher reliability.

Most model training losses were consistently reduced throughout the epochs and converged, as shown in Fig. 10. Although early training loss logs were missing for a few models (e.g., DistilBERT and ModernBERT), their final epoch losses were consistent. Given the small size of the dataset and consistent convergence behavior, we preferred direct performance indicators (ROC-AUC, confusion matrix, and accuracy) to the extra validation loss plots. Figure 10 shows the training loss for each epoch for several models trained on the hate speech detection dataset. Compared to the other models, Mistral and LLaMA-3 grew the fastest, and LLaMA-3 reached almost no loss by epoch 5. Generally, Phi−3.5 and DeepSeek achieved steady and noticeable progress in each training round.

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5 Discussion

We also studied the utility of fine-tuned large language models (LLMs) for recognizing hate speech in code-mixed Hinglish online content, and the findings are presented in this paper. We further empirically compared transformer-based architectures, such as DeepSeek R1, Gemma 3, Llama 3, Modern BERT, Distil BERT, Mistral, and Phi 3.5, and investigated their performance under different modalities (full fine-tuning and parameter-efficient regimes using adapters and low-rank adaptation). Our findings highlight the challenges of context sensitivity and adaptation faced by conventional models, which are faster to deploy than the proposed model. In contrast, the transformer-based models achieved significantly higher precision, recall, and F1 scores than the CNN-based models. DeepSeek R1 obtained the best scores for all metrics, indicating that it is resilient to the complexity of hate speech semantics, even in a code-mixed and culturally sensitive environment.

Despite these advances, there is still room for further improvement. Models such as Llama 3 had a much higher recall but much lower precision, indicating an overprediction of hate speech in borderline cases. Furthermore, scalability remains a problem, especially when users intend to execute larger models (such as Mistral) in real time. From an ethical perspective, the black-box nature of LLMs has implications for transparency, accountability, and fairness in labeling hate speech. Although CoT prompts and rational extraction approaches provide a level of interpretability, future studies should investigate more comprehensive explainability models to facilitate human supervision. Deployment trade-offs and free speech: Automated moderation reduces harm but can also over-block legitimate speech; to balance this, conservative confidence thresholds should be paired with human escalation for ambiguous or high-impact actions, transparent appeal processes and clear policy documentation. Error impact: False positives can chill speech and erode trust, whereas false negatives allow harm to persist. Operating points should be calibrated per policy (e.g., higher recall for imminent harm and higher precision for satire or borderline cases) and monitored with per-group FP/FN audits. Adversarial evasion: Classifiers are vulnerable to obfuscations (creative spellings, euphemisms, coded language) and coordinated gaming; mitigations include dynamic lexicon refresh, periodic re-tuning with hard-negative mining, and incorporating conversational context (thread, author history, parent comments) to detect implicit hate speech. Governance and oversight: Practical deployment should include bias audits, red-team testing, model cards, data provenance tracking, and regular error reviews with domain experts and community stakeholders. In high-stakes settings, models should serve as decision support rather than sole arbiters.

Finally, this evaluation emphasizes the need for balanced, high-quality, culturally sensitive, and representative datasets. Our approach to dataset formation and annotation, considering both automatic and human adjudication, was instrumental in ensuring that the model predictions were trustworthy. Future work should evaluate cross-domain robustness, simulate adversarial campaigns at scale, and study potential chilling effects via user research to refine precision–recall trade-offs and governance thresholds for different deployment contexts.

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6 Strengths of LLMs in hate speech detection

6.1 Zero-shot and few-shot learning capabilities

LLMs outperform zero-shot learning, in which they identify hate speech without task-specific training data. This is particularly beneficial in low-resource contexts or for newly emerging hate speech types. Recent LLMs have also performed well in zero-shot learning, performing comparably to fine-tuned models [20].

6.2 Contextual understanding

LLMs are good at capturing the complexities of context, which is very important for the exact identification of hate speech. Using carefully designed reasoning prompts, LLMs can leverage the knowledge acquired to achieve better performance in detecting hate speech [49].

6.3 Multilingual capabilities

LLMs generalize well to multiple languages, including low-resource languages such as Bengali. For example, GPT−3.5 Turbo has remarkably good precision in interpreting hate speech in Bengali compared with traditional approaches by a large margin [50].

6.4 Explainability and interpretability

CoT prompts and post-hoc explanations are differentiating factors that make LLMs more interpretable. It not only enhances the accuracy of the model but also produces the rationale behind the decision-making process, resulting in more interpretable and trustworthy models [28].

7 Future directions

Future work will extend the methods we propose to scale with larger models and more complicated tasks, such as open question answering and multi-hop reasoning. The construction of MT frameworks with adapters may reduce catastrophic forgetting and enable models to adapt continuously to new domains. Domain-specific LLMs for specific areas (e.g., health and finance) using domain-adaptive adapters may significantly improve the accuracy of hate speech identification within a given domain. In addition to standard model metrics, analyses should include measures of interpretability, investigations of attention, and considerations of user-facing feedback loops for direct-model predictions. The use of human feedback at different stages of the model development pipeline, such as reinforcement learning using human feedback (RLHF), active learning, and expert-guided fine-tuning, can result in a more robust, aligned, and ethical use of LLM. Such an approach is particularly attractive for delicate areas such as hate speech detection, where decisions made automatically may have significant social consequences.

8 Conclusion

Since then, LLMs have revolutionized the hate speech detection field by providing the most robust and context-aware information that was not possible previously. By leveraging modern models such as DeepSeek R1, Gemma3, Llama 3, Distil BERT, Modern BERT, Mistral and Phi 3.5, along with advanced fine-tuning techniques such as adapters and LoRa, this work achieves new state-of-the-art performance and efficiency in a custom dataset. Specifically, this study contributes (i) a balanced Hinglish dataset with

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hybrid auto-label plus expert adjudication, (ii) a consistent, transparent evaluation protocol with parameter-efficient LoRA configurations tailored per architecture, and (iii) an operational responsibility rubric spanning harm-mitigation thresholds, bias audits, and expert oversight. Future work should focus on advancing scalable, interpretable, and domain-specific approaches to mitigate the challenges of hate speech detection.

LLMs have significantly improved hate speech detection owing to their context-aware and language-unbiased understanding of hate speech. However, challenges remain in terms of bias, interpretability, and computability. Future directions include multimodal aggression and hate detection (text–image–video) for memes and shorts, pilot deployments with Indian digital platforms to quantify precision–recall trade-offs, appeal workflows, and user trust, and robustness against adversarial evasion via hard-negative mining, dynamic lexicons, and context-aware modeling.

Future research should focus on designing scalable, domain-specific, and human-aligned LLM architectures capable of processing complex linguistic structures, the context of cultural expressions, and changing manifestations of hate speech. In addition, introducing bias mitigation measures and counter-speech generation mechanisms may help shift online platforms to safer and more respectful environments.

Acknowledgements
We express our gratitude to the competent annotators who devoted their time and skills to the dataset's manual validation. This study was made possible by the Reddit platform, which provides open access to public data through its study could be possible.

Author contributions
Bhawani Singh Rathore: Conceptualization, methodology, formal analysis, investigation, resources, data curation, software, model fine-tuning, writing-original draft preparation and editing. Sandeep Chaurasia: Conceptualization, methodology, supervision, writing, review, and editing.

Funding
Open access funding provided by Manipal University Jaipur. The work received no funding.

Data availibility
This study deals with sensitive and potentially erroneous hate speech material that might be harmful in the dataset. These data are not publicly available owing to ethical considerations and platform policies. Nevertheless, the corresponding author can provide it upon reasonable request for academic and non-commercial research.

Declarations

Ethics approval and consent to participate
Not applicable. The annotators were members of the research team who participated voluntarily. No sensitive personal data were collected, and formal ethical approval was not required.

Consent for publication
Not applicable.

Informed consent
Not applicable. This study used publicly available data from Reddit without any direct interaction with individuals.

Competing interests
The authors declare no conflict of interest.

Fairness assessment
Evaluation and error analysis were stratified over demographic proxies observable in Hinglish social media (e.g., gendered terms, religion-coded terms, identity slurs). We inspected the per-group precision, recall, F1, and confusion patterns to detect disproportionate false positives/negatives. Shortcut reliance was probed with neutral identity term sentences and minimally edited counterfactuals that swap protected attributes.

Bias mitigation
The dataset was label-balanced, near-duplicate slur templates were filtered, and hard-negative examples were added iteratively. Parameter-efficient tuning enabled periodic refreshes, and high false-positive terms were flagged for targeted corrections.

Community involvement
Expert annotators applied a rubric with context-aware adjudication. For future updates, a participatory feedback loop with Indian platform moderators is planned to review FP/FN exemplars, refine thresholds for borderline categories (satire and reclaimed slurs), and document changes in a public model card.

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Deployment safeguards
Models are decision support with human escalation for ambiguous/high-impact cases; interventions are logged for audit; periodic bias audits monitor per-group error trends with predefined triggers for refresh.

Received: 17 July 2025 / Accepted: 27 October 2025
Published online: 22 December 2025

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